VibeAI: evaluating image vibe representations

Research report · 12 November 2026 · decompose-then-verify with LLM judges

Results

Three ways of writing a vibe representation were compared, all generated by gpt-5: a one-line **baseline** prompt, the guided **v1** prompt, and the **workflow**, which starts from v1 and revises it with judge feedback. Each representation is split into atomic vibe claims. **Plausibility** asks whether those claims are true of the image; **richness** asks, head to head, which system's set of vibes captures more of the image's feeling.

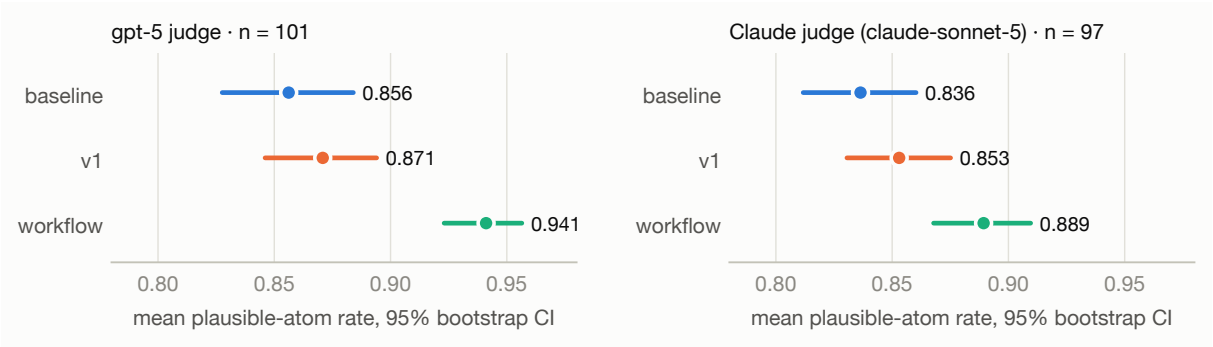


Figure 1. Plausibility. Mean share of plausible atoms per image, with 95% bootstrap intervals (10,000 resamples). Left: gpt-5 judge, 101 images. Right: independent Claude judge, 97 images all three systems share.

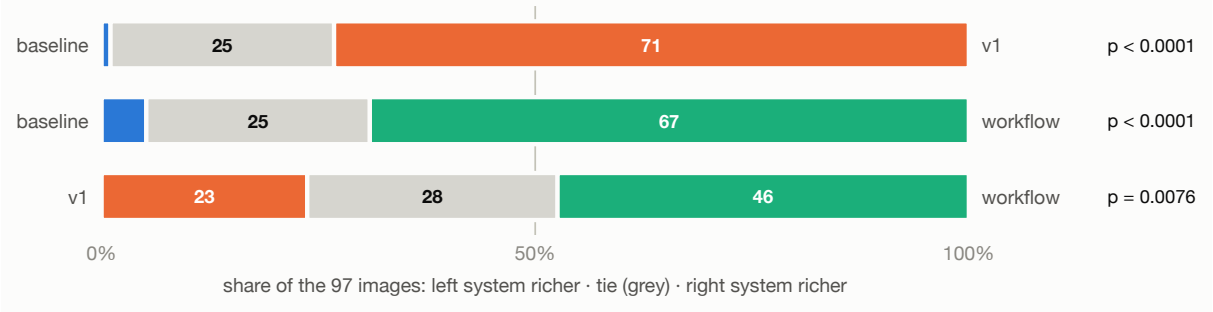


Figure 2. Richness, pairwise. For each pair, how many of the 97 images had the left system's vibe set judged richer, a tie, or the right system's richer. Claude judge, each comparison judged in both A/B orders. p is an exact two-sided sign test over the decisive (non-tie) images; all three pass Bonferroni ($p < 0.0167$).

- **The workflow makes representations more plausible, under both judges.** With the gpt-5 judge it beats baseline by +0.085 and v1 by +0.070 (both $p < 0.0001$); with the Claude judge by +0.053 ($p = 0.0005$) and +0.036 ($p = 0.0064$). Most of the gain is on evidence-backed claims (page 3).
- **v1 is not more plausible than baseline overall** ($p = 0.40$ and 0.30), although it is as good or better within each type of claim: it writes far more evidence-backed claims (63% of atoms vs 42%), the harder kind, and the shift in mix cancels the gain.
- **v1 and the workflow are far richer than baseline.** Of the decisive comparisons, v1 wins 71 of 72 and the workflow 67 of 72, and both gaps survive controlling for how many vibes each set lists (page 4).
- **The workflow's richness edge over v1 is not distinguishable from length.** It wins 46–23 with 28 ties ($p = 0.0076$), but it lists about one more vibe per image, and at equal set size the two are level ($p = 0.60$).

The gpt-5 judge shares its model with the workflow's internal gates and scores the workflow 0.051 above the Claude judge, against about 0.02 for the other systems — likely self-preference, so the Claude-judged numbers are the conservative estimate. Richness is Claude-judged only, for the same reason.

comparison	plausibility · gpt-5	plausibility · Claude	richness · Claude (wins–ties–losses)
v1 vs baseline	+0.015 (p 0.40)	+0.017 (p 0.30)	**71–25–1 ($p < 0.0001$) ✓**
workflow vs baseline	**+0.085 ($p < 0.0001$) ✓**	**+0.053 (p 0.0005) ✓**	**67–25–5 ($p < 0.0001$) ✓**
workflow vs v1	**+0.070 ($p < 0.0001$) ✓**	**+0.036 (p 0.0064) ✓**	**46–28–23 (p 0.0076) ✓** · size-controlled p 0.60

Table 1. Plausibility: mean per-image difference, two-sided paired sign-flip permutation test (20,000 resamples, seed 0). Richness: first system's wins, ties, losses over 97 images; exact sign test on decisive images. ✓ = below the Bonferroni threshold 0.0167 for three comparisons.

Pipeline

Every batch evaluation runs the same steps: generate a representation from the image, decompose it into atoms, and let LLM judges score them. The workflow closes the loop by feeding two judgements back to the generator.

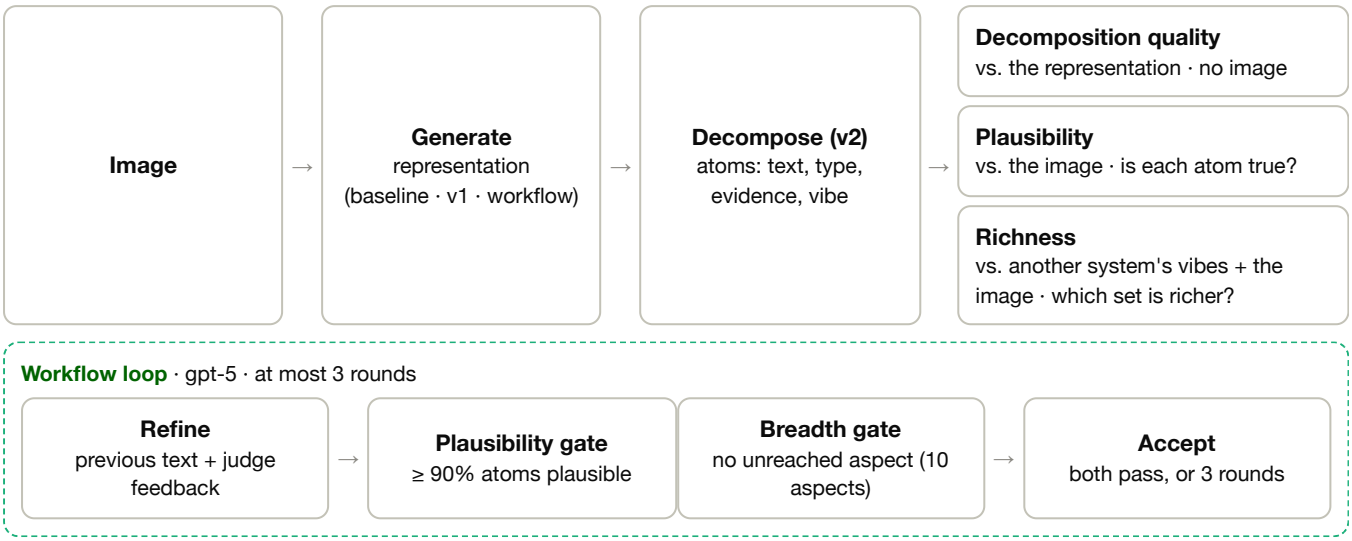


Figure 3. Batch evaluation (top) and the generator–evaluator workflow (bottom). The workflow gates on breadth, not richness: richness needs a second system to compare against and is the benchmark the workflow is scored on.

Representation prompt versions

version	what it asks for	status
baseline	One line: “Describe the vibe of this image.”	compared
v1	Flowing prose weaving evidence-backed statements (a named visual cue tied to a vibe) with vibe-only statements. Every cue must carry a vibe; vibes must be grounded and not contradicted by the image; evidence must be a main driver; no speculation.	compared
workflow	v1 as the first draft, then a refine prompt: previous text + which claims failed and which vibes are missing. Keep what passed, re-anchor or drop failed claims, add missing aspects in a few words, keep the length. The rubric is never shown to the generator.	compared
v2 · dr · rich	v2: five-step JSON self-check. dr: draft, self-question, breadth self-check, rewrite. rich: describe, imagine context, 1–3 vibe phrases.	explored

Decomposition

The v2 decomposer (used for every reported run) splits a representation into atoms that are affective, atomic, faithful to the original wording and evidence-preserving, then types each atom *vibe_only* or *evidence_backed* and extracts its evidence cues and its **vibe** verbatim. The vibe field is what the richness judge compares. The decomposition-quality judge scored the decomposer versions baseline 0.657 → v1 0.979 → v2 0.982 (101 images; v1/v2 completeness 0.99).

What each judge evaluates

judge	sees	procedure	score
Decomposition quality	representation + atoms (no image)	Completeness rated 1–5; each atom Good only if affective, atomic, faithful and evidence-preserving.	mean of completeness and 5 × share of Good atoms, ÷ 5
Plausibility	image + atoms	vibe_only: can the vibe be read off the image (any contradicting cue fails it)? evidence_backed, stopping at the first failure: is the cue present, can the vibe be read off the image, is the cue a main driver?	plausible atoms ÷ atoms
Richness	image + two systems' bare vibe sets (evidence omitted), as Set A / Set B	List the image's vibe facets; for each set, which it captures and how specifically; compare range first, then specificity. Treat every vibe as true; ignore count, length and position. Winner A, B or tie. Run in both orders: agreement or win + tie is a win; conflict or two ties is a tie.	per pair: wins, ties, losses; sign test on decisive images
Breadth (workflow gate)	image + atoms	For ten aspects (subject, interpersonal, object, environment, composition, colour and light, texture, style, activity, time): not applicable / covered / gap.	covered ÷ applicable; the gate rejects on any gap

Plausibility: where the workflow's gain comes from

A *vibe-only* atom names a feeling and passes if the image supports it. An *evidence-backed* atom also names the cue behind it and must pass three checks: the cue is present, the vibe is supported, and the cue is a main driver of the vibe. Evidence-backed atoms fail more often, so a score depends on how well each type is written and on the mix of types.

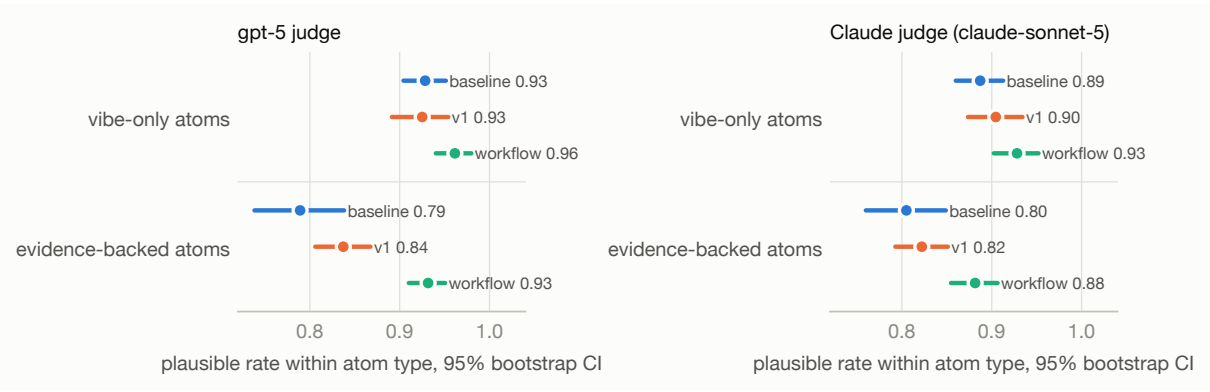


Figure 4. Plausibility by atom type. Mean per-image plausible rate within each type, with 95% bootstrap intervals. Images with no atom of a type are left out of that type.

	baseline	v1	workflow
atoms per image	9.2	11.4	12.3
share of atoms that are evidence-backed	42%	63%	69%
evidence-backed plausible rate · gpt-5 / Claude	0.79 / 0.80	0.84 / 0.82	0.93 / 0.88
vibe-only plausible rate · gpt-5 / Claude	0.93 / 0.89	0.93 / 0.90	0.96 / 0.93
gpt-5 judge – Claude judge (97 shared images)	+0.018	+0.023	+0.051

Table 2. Atom mix and per-type rates. Atom counts and mix are from the gpt-5-judged runs (101 images); the Claude-judged runs cover 99–101 images and their mix differs by at most 0.03.

Reading it. v1's guidance raises evidence-backed accuracy (0.79 → 0.84 under gpt-5) and leaves vibe-only accuracy level, but it also moves a fifth of all atoms from the easy type to the hard one. The two effects cancel, which is why v1's overall score barely moves. The workflow keeps v1's evidence-heavy mix and lifts the hard type to 0.93 (0.88 under Claude). The refine step is told which claim failed and why. Of the 150 first-draft failures, 77 were feelings the image works against, which a rewrite drops or qualifies; 62 were mapping failures — a real cue tied to the wrong feeling — which it can fix by naming the cue that actually drives the vibe; 11 named a cue that isn't there.

What the workflow loop did

over 101 images, gpt-5 throughout	first draft	final
plausible-atom rate (loop's own judge)	0.871	0.941
implausible atoms per image	1.49	0.70
breadth gaps per image (vibe aspects left unnamed)	1.38	0.23
atoms per image	11.4	12.3

Table 3. 73% of images passed both gates within the 3-round limit (mean 2.4 rounds; 12 accepted on the first draft). 21 images ended below the best round they passed through, since the loop returns its last round, not its best. The added atoms likely come from the breadth gate, which asks for missing vibes; that is where the richness check on page 4 picks up a length difference.

Example: one implausible atom, fixed in the loop (city_tokyo.jpg, plausibility 0.875 → 0.80 → 1.00)

First draft
"The tangle of overhead wires and close, weathered facades create an intimate, slightly nostalgic urban cocoon, more whisper than bustle."
Mapping check failed: the wires shape the intimacy, but the quiet comes from the low activity and the hour.

Final round
"Scattered pedestrians read as soft silhouettes with space between them, and that spacing lets the quiet breathe. Overhead wires and close, weathered facades wrap the view into an intimate, slightly nostalgic urban cocoon."
Same vibes, each tied to the cue that drives it; 8 → 10 atoms.

Richness: is a richer set just a longer one?

The judge is told not to count vibes, yet the system ranking follows set size: the workflow lists 12.3 vibes per image, v1 11.4, baseline 9.2 (the workflow's extra vibes likely come from its breadth gate, page 3). To separate the two, the decisive comparisons were fitted with a Bradley–Terry model — logistic regression on pairwise outcomes, the analogue of ANCOVA for wins and losses — with the difference in set size as a covariate:

$$\text{logit } P(x \text{ judged richer than } y) = (\theta_x - \theta_y) + \beta \cdot (\text{size}_x - \text{size}_y)$$

θ is a system's richness at equal set size (baseline fixed at 0); β is what one extra vibe is worth.

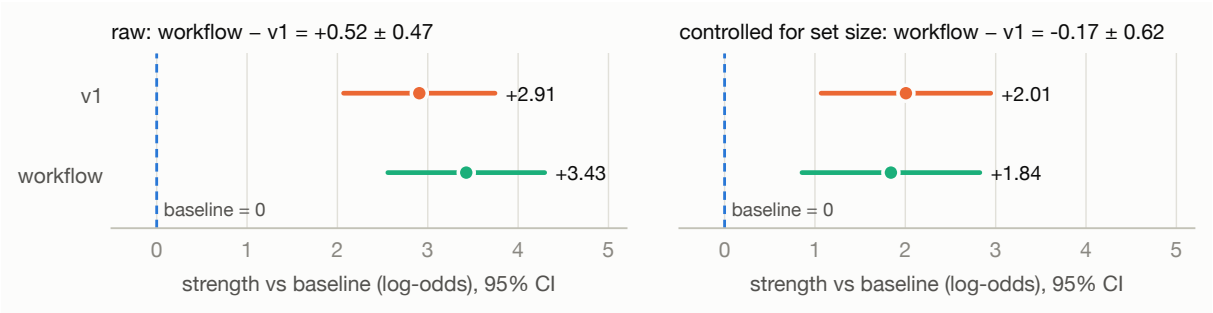


Figure 5. Richness strength vs baseline, raw (left) and with set size as a covariate (right), with 95% Wald intervals. 213 decisive comparisons; fitted with `vibeai.eval.richness_control` and cross-checked against statsmodels.

term	raw	controlled for set size
v1 vs baseline	+2.91 (odds ×18), p < 0.0001	+2.01 (odds ×7.4), p < 0.0001
workflow vs baseline	+3.43 (odds ×31), p < 0.0001	+1.84 (odds ×6.3), p = 0.0002
workflow – v1	+0.52 (odds ×1.7), p = 0.030	–0.17 (odds ×0.85), p = 0.60
each extra vibe (β)	—	+0.72 (odds ×2.1), p < 0.0001

Table 4. Bradley–Terry coefficients (log-odds of being judged richer). Adding the size term improves the fit: likelihood-ratio $\chi^2(1) = 37.0$, $p < 0.0001$. The raw workflow – v1 p (0.030) differs from the sign test's (0.0076) because the model pools all three pairs.

Reading it. v1 and the workflow beat baseline even at equal set size — about 6–7× the odds of being judged richer — so that gap is not length. The workflow's gain over v1 disappears once size is in the model. That does not prove the workflow is no richer: if its extra vibes really reach new facets, part of a true advantage runs through size and the control removes it too. What it does say is that the data cannot tell "covers more" from "lists more", and with each extra vibe roughly doubling the odds of a win, a length bias in the judge is a live possibility. A blind human comparison on size-mismatched pairs is what would settle it.

Judge diagnostics

diagnostic	value	meaning
order consistency	69.8% (72 / 71 / 66% by pair)	share of comparisons whose two orders reached the same verdict; the rest are scored as ties
Set A win rate, single calls	195 / 530 = 36.8%, p < 0.0001	the judge prefers whichever set it reads second; judging both orders cancels this per comparison
calls · errors · cost	582 · 0 · ≈ $6	2 orders × 3 pairs × 97 images; ~2,000 input and ~580 output tokens per call

Table 5. The position bias is strong enough that a single-order design (even randomised) would have added a large amount of noise to every verdict.

Example of an order conflict (v1 vs workflow, 1759287348909038.jpg — a person reading in a car under an open sunroof)

v1 · 11 vibes
airy, free, private hideaway, slow-living comfort, cocooned comfort, retreat, a quiet pause tucked into a journey, gentle calm, unhurried road-trip, wide world outside, mind blissfully elsewhere

workflow · 8 vibes
the real sense of openness—air and landscape pouring in, mid-afternoon mood, deliberate pause, road-trip feeling, wide-world-outside feeling, simply softens the edges, candid ease, lifestyle ease

In both orders the judge picked the set it read second, citing the workflow's light, time-of-day and candid facets one time and v1's cocooned, private-sanctuary feeling the other. The verdict follows position, so it is scored a tie.

How the judges were validated

A human annotator labelled the same items as the judge in a purpose-built web app; agreement is Cohen's κ, which discounts the agreement expected by chance.

judge · labels	items	κ	raw agreement	reading
Plausibility · atom plausible?	268 atoms / 30 images	0.817	95.9%	almost perfect
↳ evidence_backed atoms	102	0.888	96.1%	
↳ vibe_only atoms	166	0.673	95.8%	substantial
Decomposition · completeness (1–5)	30 images	0.696 linear-wtd	83.3% exact	substantial
Decomposition · atom Good/Bad	339 atoms	0.644	82.3%	substantial
↳ fidelity · evidence preservation	339	0.626 · 0.594	82.6% · 79.6%	
↳ atomicity · affectiveness	339	0.305 · 0.000	96.2% · 99.7%	base-rate effect*
Richness · which set is richer?	—	—	—	not yet validated

* Almost every atom was atomic and affective, so chance agreement is near 100% and κ collapses even though judge and human disagree on only 1–13 atoms. Decomposition κ recomputed for this report. Plausibility κ is carried over from the previous report: it was measured on the gpt-5 judge with the baseline representation and v1 decomposition, under an earlier output format that the current agreement script no longer reads (prompt logic unchanged). Richness, breadth and the Claude judges have not been human-validated; the web app now has a blind pairwise richness annotation page for that.

Method and limitations

- **Significance.** Plausibility: paired sign-flip Monte Carlo permutation test on per-image differences, two-sided, 20,000 resamples, Bonferroni across three comparisons (`results/significance/`). Richness: exact two-sided binomial sign test on decisive comparisons, same Bonferroni threshold.
- **Richness is relative.** It ranks the compared systems against each other; it says nothing about how much of an image's vibe any of them captures in absolute terms, and adding a system means judging new pairs.
- **Set size.** The workflow-vs-v1 richness gap cannot be told apart from set length (page 4). The judge prompt forbids counting, but the size effect is large.
- **Position bias.** The judge favours the set it reads second (63% of decisive single calls). Both-order judging cancels it per comparison, at the cost of turning about 30% of comparisons into ties.
- **One judge, one annotator.** Richness uses one Claude model, chosen to avoid gpt self-preference on gpt-5 outputs; it has not been checked against a human or a second judge. All agreement figures come from one annotator.
- **Image sets.** Plausibility uses all 101 main images (gpt-5 judge) or the 97 every system completed under the Claude judge; richness uses the same 97.
- **Retired: pool-based richness.** The previous report scored richness as coverage of a 4-branch "vibe pool" on 10 holdout images and found no significant differences. That method was dropped: the pool depended on its construction prompts and an LLM synonym merge that could not be verified, and its cost limited it to 10 images. Its data is archived outside the repository; its numbers should no longer be cited.